torch.linalg.matrix_exp#

https://pytorch.org/docs/stable/generated/torch.linalg.matrix_exp.html#torch

Description:

Computes the matrix exponential of a square matrix. Letting K\mathbb{K}K be R\mathbb{R}R or C\mathbb{C}C, this function computes the matrix exponential of A∈Kn×nA \in \mathbb{K}^{n \times n}A∈Kn×n, which is defined as If the matrix AAA has eigenvalues λi∈C\lambda_i \in \mathbb{C}λi∈C, the matrix matrix_exp(A)\mathrm{matrix_exp}(A)matrix_exp(A) has eigenvalues eλi∈Ce^{\lambda_i} \in \mathbb{C}eλi∈C. Supports input of bfloat16, float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions.

Function Signature

```
torch.linalg.matrix_exp(A) → Tensor#
Parameters
```

Code Examples

```
>>> A = torch.empty(2, 2, 2)
>>> A[0, :, :] = torch.eye(2, 2)
>>> A[1, :, :] = 2 * torch.eye(2, 2)
>>> A
tensor([[[1., 0.],
         [0., 1.]],

        [[2., 0.],
         [0., 2.]]])
>>> torch.linalg.matrix_exp(A)
tensor([[[2.7183, 0.0000],
         [0.0000, 2.7183]],

         [[7.3891, 0.0000],
          [0.0000, 7.3891]]])

>>> import math
>>> A = torch.tensor([[0, math.pi/3], [-math.pi/3, 0]]) # A is
skew-symmetric
>>> torch.linalg.matrix_exp(A) # matrix_exp(A) = [[cos(pi/3),
sin(pi/3)], [-sin(pi/3), cos(pi/3)]]
tensor([[ 0.5000,  0.8660],
        [-0.8660,  0.5000]])
```

Detailed Documentation

Documentation

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A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions.